Proof-of-concept handover: grounded document assistant

Sample of what a Starter engagement delivers. Northwind Logistics is a fictional company and every document, person and figure in the demo corpus is invented.

1. What was built

A question-answering assistant over a private document set that:

- answers **only** from the documents, quoting the exact passage for every sentence;
- says "not found" instead of guessing;
- shows each user only what their group may read (HR sees HR files, Engineering sees engineering files);
- opens the cited PDF page with the quoted text highlighted.

Piece	Choice
Sources	File upload (PDF, DOCX, HTML, text) and a polite, SSRF-guarded web crawl
Search	Postgres + pgvector + full-text, fused with RRF, local cross-encoder rerank
Access control	Keycloak single sign-on; groups become an ACL filter **inside the SQL query**
Answering	OpenAI or Anthropic via LiteLLM; a validator checks every quote against the source
UI	Next.js chat with a citation panel
Deployment	`docker compose` locally; Terraform for Azure (Container Apps, Postgres, Key Vault)

2. Results

Everything between the markers is generated from the newest report in `eval/reports/` by `make docs`.

Generated from `eval/reports/2026-10-26_7c1117f.json` by `make docs`; do not edit by hand.

Run `20261026T074347Z-370aa5`, mode `llm`, 30 questions (20 answerable / 6 unanswerable / 4 acl-denied), git `7c1117f`, embeddings `BAAI/bge-small-en-v1.5`, reranker `BAAI/bge-reranker-base`.

Environment: Python 3.12.14, torch 2.14.0, device `mps`.

Metric	Result
Retrieval recall@6, file level	19/20 (0.95) [95% CI 0.76–0.99]
Retrieval recall@6, page level (PDF)	11/12 (0.92) [95% CI 0.65–0.99]

Metric	Result
Chunks returned that the asking user may not read	**0** of 180
Forbidden documents retrieved (ACL-denied questions)	**0**
Answer correctness (macro fact fraction)	0.95 over 20 answers
False abstention	0/20 (0.00) [95% CI 0.00–0.16]
Abstention accuracy, unanswerable + ACL-denied	10/10 (1.00) [95% CI 0.72–1.00]
Citation precision	20/20 (1.00) [95% CI 0.84–1.00]
Latency p50 / p95	1,316 ms / 2,392 ms

SPEC §9.4 target	Goal	Result	Status
Retrieval recall@6 (file level)	>= 0.90	19/20 (0.95) [95% CI 0.76–0.99]	not demonstrable at this n
Abstention accuracy (unanswerable + ACL-denied)	>= 0.95	10/10 (1.00) [95% CI 0.72–1.00]	not demonstrable at this n
False abstention rate	<= 0.10	0/20 (0.00) [95% CI 0.00–0.16]	not demonstrable at this n
Citation precision	>= 0.90	20/20 (1.00) [95% CI 0.84–1.00]	not demonstrable at this n

Status uses the 95% Wilson interval, not the point estimate: at 30 questions most targets cannot be demonstrated either way.

Report files under `eval/reports/` include the per-question table so every number can be traced.

3. How to run it

```
cp .env.example .env        # placeholders only; add ANTHROPIC_API_KEY or
        OPENAI_API_KEY to enable answers
make up                     # Postgres, Redis, Keycloak, API, worker, web
make ingest-demo            # index the 100 fictional demo documents
```

Open http://localhost:3000 and sign in as `alice` (`alice-demo-only`) or `bob` (`bob-demo-only`). Ask both "What is the salary band for L5 engineers?": Alice gets a cited answer, Bob gets "not found". These accounts exist only in the throwaway local Keycloak.

- Tests: `make test` · Lint and types: `make lint` · Eval: `make eval EVAL_ARGS=--yes` (needs a key)

4. What is not proven yet

- **Answer quality with a live model** is unmeasured until the eval is run with your key.

- **The Azure deployment** passes validation, unit tests and a security scan but was never applied.
- Only 30 evaluation questions exist, so confidence intervals are wide.
- Entra ID sign-in, additional connectors (Drive, SharePoint, databases) and dashboards are not built.

5. Suggested next steps for a real engagement

1. Point the pipeline at a sample of your real documents and write 60+ questions with your domain experts.
2. Run the eval with your chosen model, tune the abstention threshold on a dev split, keep a hold-out.
3. Map your identity provider's groups to collections (Entra ID support needs a small verifier change).
4. Apply the Terraform in a sandbox subscription, then add monitoring and a retention policy.
5. Decide on rate limits, audit-log retention and who administers collections.